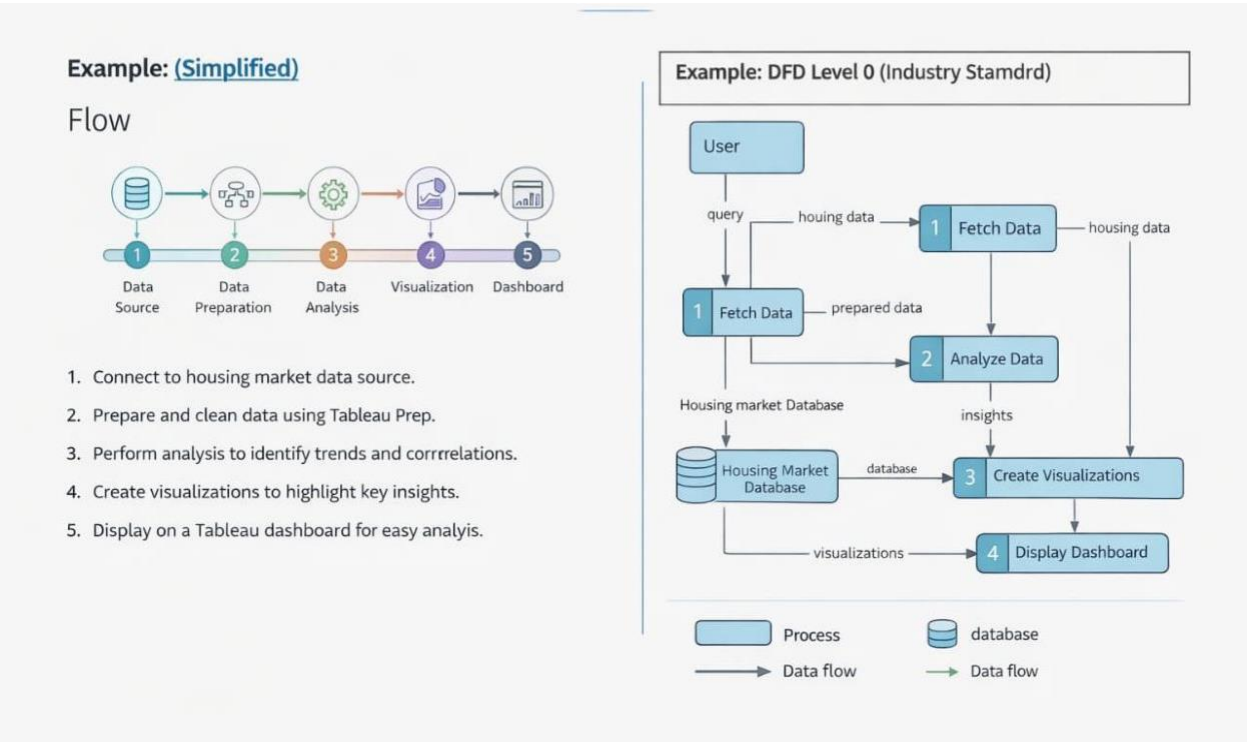
Requirement Analysis

Data Flow Diagram and User Stories

Team Id	SBAP003097
Project Name	Visualizing housing market trends: Analysis of sales prices feature using tableau

Data Flow Diagram

The Data Flow Diagram (DFD) explains how housing data moves from collection to visualization. First, raw data such as sale prices, property features, and location details is collected from databases or datasets. This data is then cleaned and processed to remove errors and inconsistencies. After preprocessing, the structured data is stored securely and connected to Tableau for analysis. Tableau generates interactive dashboards and visualizations like charts and graphs. Finally, users such as analysts and investors interact with these dashboards to gain insights and make informed decisions.



User Stories

User Type	Functional Requirements (Epic)	User story number	User story/Task	Acceptance criteria	Priority	Release
Data Analyst	Data Connection	USN-1	As a data analyst, I can connect Tableau to housing market datasets (CSV/Excel/Database) to begin analysis.	Data source connects successfully and preview is visible in Tableau	High	Sprint-1
Data Analyst	Data Preparation	USN-2	As a data analyst, I can clean and transform housing data using Tableau Prep to ensure data quality.	Missing values handled, data types corrected, duplicates removed.	High	Sprint-1
Data Analyst	Data Integration	USN-3	As a data analyst, I can join multiple datasets (sales, location, property features) for comprehensive analysis.	Datasets are joined correctly and relationships validated.	High	Sprint-1
Data Analyst	Feature Engineering	USN-4	As a data analyst, I can create calculated fields (price per sqft, yearly growth rate).	Calculated fields produce accurate values	Medium	Sprint-2
Business Analyst	Trend Analysis	USN-5	As a business analyst, I can analyze sale price trends over time to identify market patterns.	Time-series charts display accurate trends by year/month.	High	Sprint-2
Business Analyst	Feature Comparison	USN-6	As a business analyst, I can compare property features (bedrooms, bathrooms, area) against sale prices.	Scatter plots/bar charts correctly show feature-price	High	Sprint-2

				relationships		
Business Analyst	Geographic Analysis	USN-7	As a business analyst, I can visualize housing prices by location using maps.	Interactive map displays prices by region/city	High	Sprint-2
End User	Dashboard Creation	USN-8	As a stakeholder, I can view an interactive dashboard summarizing key housing market insights.	Dashboard contains filters, KPIs, and interactive charts	High	Sprint-3
End User	Filtering and Drill-down	USN-9	As a stakeholder, I can filter data by year, location, and property type.	Filters update all visualizations dynamically.	High	Sprint-3
End User	Reporting	USN-10	As a stakeholder, I can export dashboards and reports to PDF or image format.	Reports export successfully without data loss.	Medium	Sprint-3
Administrator	Data Refresh	USN-11	As an administrator, I can schedule automatic data refresh in Tableau.	Data refresh runs at scheduled intervals without errors	Medium	Sprint-3
Administrator	Access Control	USN-12	As an administrator, I can manage user access and permissions.	Users can only access dashboards based on assigned roles.	High	Sprint-3